Grade 11 – Term 1
Choosing a Statistical Model: C10 Practice

1 Your Assignment

Write an **individual PDF article analyzing a realistic situation using probability and statistics studied in Term 1**. Identify a question, select an appropriate model, apply relevant procedures, and explain what the results mean. Connect concepts from C1–C9 as appropriate to your question; you do not need to use every distribution or statistical method.

The main purpose of C10 is to select and justify an appropriate mathematical model, not simply to perform calculations. Your reasoning must connect the model's assumptions to the characteristics of the situation and the information available. Use methods already studied during the term.

2 Prepare Your Analysis

1. **Choose a focused real-life situation.** Possible starting points include delivery or waiting times, variation in household spending, business costs, or the timing of events within a repeating cycle. Choose a situation for which observations or credible contextual information can be obtained. State a specific question about a probability, typical value, variability, or standardized sample mean. A broad topic such as "business statistics" is not sufficient. Do not copy a practice problem and change its names or numbers.
2. **Define the context and variables.** Explain the purpose of the analysis and who could use the findings. Define your random variable, units, support (possible values), and relevant time period or population. State whether it is discrete or continuous. Distinguish an individual observation from a sample mean whenever this matters to your model.
3. **Organize and identify your information.** Present relevant observations or source data in a clear table, frequency table, or histogram as appropriate. State the source, collection method, sample size, and units. Clearly distinguish **real observations, theoretical assumptions, estimated quantities, and simulated data**. Explain how estimates were obtained. If you generate a sample, identify the distribution, parameters, sample size, and tool used; do not describe generated values as real observations. Simulation illustrates the assumed model and does not independently establish that it fits reality.
4. **Select and justify your model.** Name the distribution or statistical approach and specify its parameters, explaining whether these are known, estimated, or assumed. Support your choice using the variable's support, shape, data pattern, and relevant conditions, such as a plausible constant event rate or independent observations. Briefly explain why a plausible alternative studied this term is less suitable. Missing information must be acknowledged; a convenient formula alone does not justify a model.
   Use the following connections to the term's materials to identify a suitable approach:
   - **C1–C3:** Characterize the variable and its probability model. For continuous uniform, linear, triangular, or piecewise models, construct a valid density and calculate relevant interval or tail probabilities using geometric areas. For a circular variable, define one cycle and its cut, and split any interval that crosses the cut.
   - **C4–C5:** For a justified constant-rate waiting-time model, use exponential probabilities. Where relevant, calculate and interpret a continuous distribution's theoretical mean, variance, and standard deviation using the formulas studied.
   - **C6–C7:** Use relative frequencies to estimate interval probabilities and compare them with theoretical probabilities. Calculate sample means and sample variances and compare them with the model's corresponding values. Label approximations from grouped data and explain the role of sample size and sampling variation.

- **C8–C9:** For standardized sample-mean analysis, justify standard normal or Student's t using whether the population standard deviation is known or estimated. State the independent random-sample and normal-population assumptions used in the practice materials; for t, give $\nu = n - 1$. Use appropriate one-tail, two-tail, or central areas and critical values. A large sample does not make an unknown population standard deviation known.

These are options, not a checklist of required methods. Choose a coherent combination that answers your question. Student's t models a standardized sample mean under the stated assumptions; do not select it for individual values merely because your sample is small.

5. **Carry out the relevant procedures.** State the event or quantity being calculated, show the formula or geometric method, substitute your values, and report the result. Include only calculations that support your question. For a constructed density, check nonnegativity and total area 1. For sample variance, use the $n - 1$ denominator studied in C7. When using probability tables, identify the table convention and degrees of freedom where applicable; report table-based approximations or bounds rather than unsupported exact probabilities. Label graph axes, units, and any shaded probability region.
6. **Interpret and evaluate the findings.** Explain each main result in the original context, distinguishing theoretical probabilities from observed relative frequencies and standardized cutoffs from quantities in original units. Discuss limitations such as sample size, unrepresentative observations, uncertain parameters, or assumptions that may fail. Explain how these affect your conclusions. State what the evidence supports and avoid treating model predictions as guaranteed outcomes.

3 Required Article Structure

Write a connected academic article with the following components. Integrate tables, graphs, and calculations into the discussion; do not submit disconnected answers to the preparation steps.

Title and Author Give a descriptive title and your full name.

Introduction Present the real-life situation, question, and objective of the analysis.

Methodology Define variables, data sources, sample size where applicable, assumptions, estimates, and any simulation. Specify and justify the model and selected procedures.

Results and Discussion Present the mathematical analysis, necessary calculations, tables or graphs, contextual interpretation, and limitations.

Conclusion State the main findings and their implications, supported by your analysis.

References List the sources of contextual information, data, and any tools used. Give enough detail to locate each source (author or organization, title, date, and link where available), and cite sources where used. Identify your own data collection explicitly.

4 Before You Submit

Submit a **readable, professionally organized PDF article**. Use clear headings, complete paragraphs, consistent notation, and legible tables and graphs. Check that the model choice is justified, all information is correctly labeled by origin, calculations answer your question, and conclusions acknowledge limitations. Open the exported PDF to confirm that equations, tables, and graphs are complete and readable.